

RESEARCH ARTICLE

Marine Fouling Release Performance of Amphiphilic Carboxybetaine Perfluoropolyether Methacrylate Polymer Coatings

Florian Koschitzki, Onur Özcan, Robin Wanka, Marc Krisam, and Axel Rosenhahn*

Amphiphilic polymer systems have been demonstrated to be an effective and environmentally friendly approach to combat the challenges of biofouling, especially in the real ocean environment. A variety of polymer compositions containing fluorophilic perfluoropolyether (PFPE) urethane dimethacrylate and zwitterionic carboxybetaine methacrylate (CBMA) are in situ copolymerized across a broad concentration range and anchored to chemically functionalized substrates by grafting-through polymerization. Dynamic accumulation assays with the marine diatom *Navicula perminuta* and the marine bacterium *Cobetia marina* revealed an enhanced antifouling performance in most of the amphiphilic polymer mixtures when zwitterionic units are combined with hydrophobic PFPE compounds. It is noticed that the resulting polymer networks still possess enough orientational freedom to quickly rearrange in the presence of water. It is further observed that several of the very hydrophilic formulations and some of the moderately hydrophobic formulations exhibit low fouling against the marine fouling organisms *Navicula perminuta* and *Cobetia marina*. This correlated with increased surface roughness and the ability for underwater rearrangement. Furthermore, it is discussed whether the probability that highly hydrated hydrogels, which appear sponge-like, may create a diffuse interphase which promotes the incorporation of silt. Latter is counteracted by the hydrophobic PFPE building blocks in the amphiphilic coatings.

1. Introduction

Conventional approaches to counteract the undesired accumulation of biomass on submerged surfaces, widely known as biofouling, mainly rely on biocides released by an ablative or leaching process.^[1,2] As the use of biocides is increasingly restricted

by legislation, the research on environmentally benign solutions such as fouling-release (FR) technologies gained popularity.^[1,3,4] This technology enables the removal process of already attached marine organisms simply by the motion of a vessel through the water. Mechanical material properties, surface energy, and the roughness on top of the coatings are proposed to be dominant factors in resisting the adhesion of fouling organisms.^[5,6] Generally, the design of smooth surfaces with low surface energy seems advantageous to facilitate fouling release by avoiding strong polar interactions such as ionic or hydrogen bonding between the surface and the biomolecules. The weaker the interactions are, the easier the organisms can be removed by the motion of the fouled vessel in the water and the associated hydrodynamic forces acting on the hull.^[7,8] In this context, elastic materials like fluorinated polymers or silicones with low surface energy and low Young's modulus were identified as suitable candidates and have been successfully applied as marine FR coatings.^[9–14] As

a recently emerging class of materials with good FR properties, hydrophilic and hydrophobic components have been merged into so-called chemically “ambiguous” surfaces.^[15–20] These amphiphilic systems exhibit high resistance against non-specific protein adsorption as well as against fouling organisms such as *Navicula incerta* and *Ulva* spores. Typically, hydrophilic polyethylene glycol (PEG) units are blended into fluorinated polymers.^[13,16,21–26] Alongside a low surface energy, hydrophobicity, and oleophobicity, fluoropolymers also present strong chemical and thermal stability which makes them an attractive component for the use in amphiphilic networks.^[17] PEGylated substances, on the other side, provide long and flexible hydrophilic chains and with their highly hydrated interfaces, they represent well-studied antifouling components known for effective inhibition of protein and cell adsorption.^[27–30] The enhancement of the FR performance toward both *Ulva* sporelings and diatoms by combining these two substance classes in hyperbranched or fluoroalkyl block copolymers is very well documented.^[16,21–25,31]

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While PEGylated materials are extremely interesting low fouling materials, they suffer from oxidation.^[26]

Among alternative candidates for ultralow fouling, zwitterionic chemistries have gained significant attention in recent studies.^[32–40] Unlike PEG, zwitterionic compounds carry a negative and positive charge and thus form hydration layers via electrostatic interactions that allow them to bind large amounts of water, which is proposed to be the key mechanism for their ultralow fouling properties.^[41,42] As a result, the thermodynamic gain from the adhesion of organisms and proteins is lowered and the energetically more favorable bound water leads to a high protein and cell resistance.^[37,42,43] Zwitterionic moieties are effectively counteracting adhesion, no matter if applied as self-assembled monolayers (SAM),^[44–46] as betaine polymer brushes,^[35] or as thin polymer films.^[33,34,47,48] Zwitterionic coatings effectively counteract biofilm formation,^[49] for example of the pathogenic organisms *Pseudomonas aeruginosa*,^[50] and *Candida albicans*,^[51] as well as fresh- and seawater organisms such as diatoms, e.g. *Navicula incerta*,^[52] *Ulva* sporelings,^[45] and barnacles, namely *Balanus improvisus* and *Balanus amphirite*.^[53]

Perfluoropolyethers (PFPE) are typically used as lubricants as they are viscous oils. Structurally they consist of fluorinated polyether chains, which is why they exhibit excellent thermal, photo, and chemical stability together with low surface tensions, low friction coefficients, and water repellency.^[54,55] Bioinspired by the *Nepenthes* pitcher plant and as an evolution of the lotus effect,^[56] rough solid surfaces can be protected with an infused lubricant yielding so-called slippery liquid-infused porous surfaces (SLIPS).^[57,58] Since the protective lubricant shows immiscibility with aqueous media, SLIPS have been effectively applied in the antifouling sector to prevent bacterial adhesion and reduce biofouling.^[59] The SLIPS approach has also proven to be an effective concept against marine fouling.^[58,60,61] Infusing PFPE oils into PFPE-containing polymers based on urethane methacrylate chemistry has been reported to result in a fluoropolymer matrix with tunable porosity, that proved to successfully reduce the biofilm growth of *Pseudomonas aeruginosa*.^[59] The unique low fouling properties of PFPE, even if used as a polymer without any oil, were discovered early in a pioneering study in which amphiphilic polymers were obtained by copolymerizing PFPE urethane methacrylates with PEG-methacrylates to form low fouling marine coatings.^[13,62] The immiscibility of fluoropolymers with non-fluorinated compounds impedes preparation and can require pre-assembly of copolymers to avoid phase separation and bypass poor adhesion to the supporting matrix.^[63–66] In a very recent investigation, zwitterionic sulfobetaine methacrylate, as an alternative to PEG, was copolymerized with PFPE urethane dimethacrylates using a sandwich-assembly.^[53] When these amphiphilic coatings were tested against the diatom *Navicula incerta*, two different barnacle species, and the bacterium *Staphylococcus aureus*, the results revealed a strong reduction of fouling when zwitterions were integrated into the PFPE matrix.^[53] Although the zwitterionic content clearly enhanced the performance of the coatings, a strong contribution to the roughness of the tested coatings was discussed. PFPE containing amphiphilic coatings appears to be an interesting class of materials for marine applications with a high demand for further investigations.

Here we report the facile in situ copolymerization of zwitterionic carboxybetaine methacrylate (CBMA) with PFPE ure-

thane dimethacrylate in the direct vicinity of methacrylate-functionalized glass substrates. The work follows a published concept in which very smooth and homogeneous amphiphilic coatings based on ethylene glycol dicyclopentenyl ether acrylate (DCPEA) and CBMA were successfully prepared and tested against different marine fouling species.^[67] Using this method, nine different amphiphilic CBMA-PFPE methacrylates were produced by grafting-through polymerization and compared to hydrophilic CBMA and hydrophobic PFPE polymers. All coatings were characterized and analyzed regarding their marine fouling resistance against *Navicula perminuta* and *Cobetia marina*.

2. Results

Free-radical polymerization was used to in situ copolymerize PFPE urethane dimethacrylate and zwitterionic CBMA by grafting-through mechanisms in the presence of ethylene glycol diacrylate (EGDA) as crosslinker and 2,2-dimethoxy-1,2-diphenylethane-1-one (DMPAP) as photoinitiator (**Figure 1A**). To overcome the immiscibility of the two chemically incompatible monomers, a mixture of 2,2,2-trifluoroethanol (TFE) and 1H,1H,2H,2H-perfluoro-1-octanol (PFO) was used as a solvent. For covalent coupling of the polymer coatings, the substrate was pre-modified with a methacrylate-terminated silane as an anchor group. The polymerization was carried out under controlled airtight conditions to prevent solvent evaporation and oxygen diffusion, occurrences that were found to quench the polymerization and/or lead to phase separation. The UV irradiation times were adjusted for each monomer composition since different polymerization rates were observed for the hydrophilic CBMA compared to its hydrophobic counterpart. The solvent mixture of TFE and PFO was also selected according to the monomer ratio, while the overall solvent-to-monomer weight ratio remained constant. Too low polymer concentrations resulted in inadequately rough and opaque instead of ultra-smooth and transparent polymer coatings (**Figure 1C**). This observation is in accordance with the findings of Keller et al., where tunable porosity of the coating was achieved by choice and the amount of solvent used.^[59] Polymers with high contents of the hydrophobic PFPE appeared hard and comparably brittle, while high concentrations of the hydrophilic CBMA exhibited rather soft and sticky properties. It is noteworthy that not only the amphiphilicity but also the degree of crosslinking is changed by the incorporation of higher concentrations of zwitterions, as it lowers the relative content of crosslinkable PFPE, in turn. The nomenclature for the obtained amphiphilic polymers follows the concept of Ruiz-Sanchez et al. with the notation xFyZ, with F as in Fluorolink MD700 and Z as in Zwitterion, while x and y represent the weight ratios of the individual components.^[53] The decision to use weight ratios as opposed to stoichiometric amounts was based upon calculations, in which the ratio of the functional moieties from tetrafluoroethyleneoxide/difluoromethyleneoxide to CBMA was best reflected by the weight ratios.

The presence of both compounds in the CBMA/PFPE copolymer can be verified by ATR-FTIR analysis (**Figure 2A**). The IR spectra of all polymers were normalized to the carbonyl stretching vibration $\nu(\text{C=O})$ at 1725 cm^{-1} . Typical vibrational modes of the hydrophobic PFPE compound were $\nu(\text{N-H}) \approx 3355\text{ cm}^{-1}$, $\nu(\text{CN})$ at 1227 cm^{-1} , and $\nu(\text{C-F})$ at 1055 cm^{-1} . The

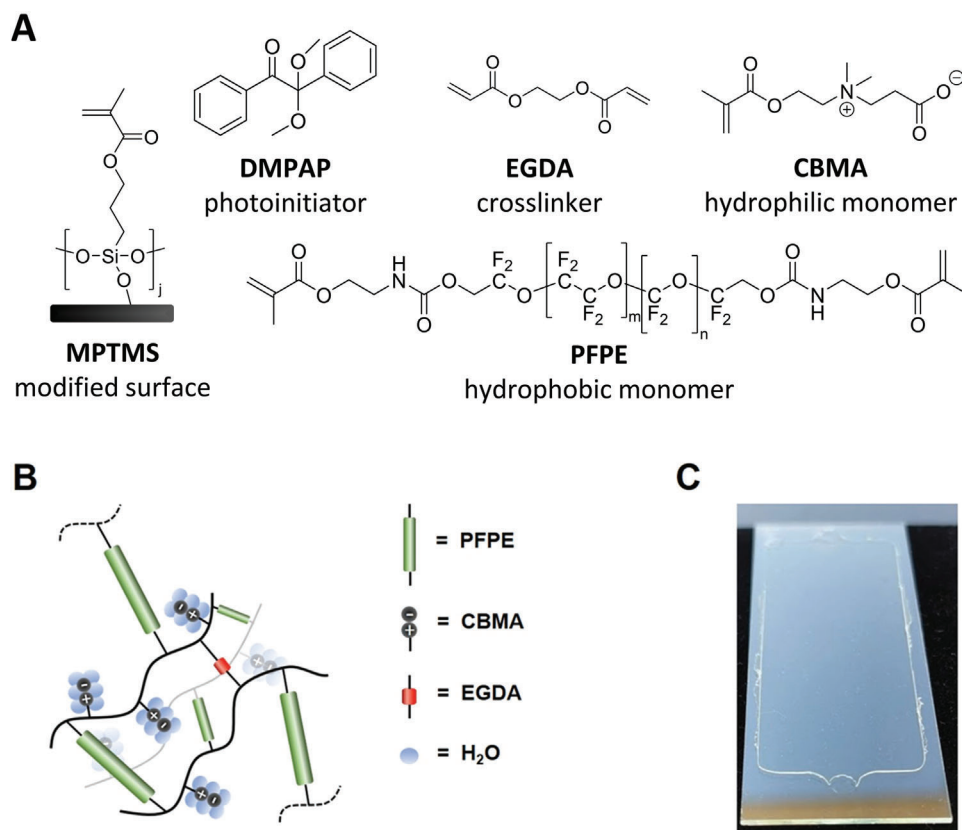


Figure 1. A) Chemical structures and abbreviations of the compounds used for the amphiphilic polymer coatings consisting of different weight ratios of the hydrophobic PFPE urethane dimethacrylate (with $M \approx 2000 \text{ g mol}^{-1}$) and the hydrophilic CBMA. 3.0 wt.% EGDA acting as crosslinker and 1.0 wt.% DMPAP as photoinitiator were used. The glass substrates were silanized with 3-(trimethoxysilyl)propyl methacrylate (MPTMS) to tether the polymer. B) Schematic representation of crosslinked amphiphilic polymer chains. C) Exemplary image of synthesized polymers: polymer coating tethered to an MPTMS-modified substrate.

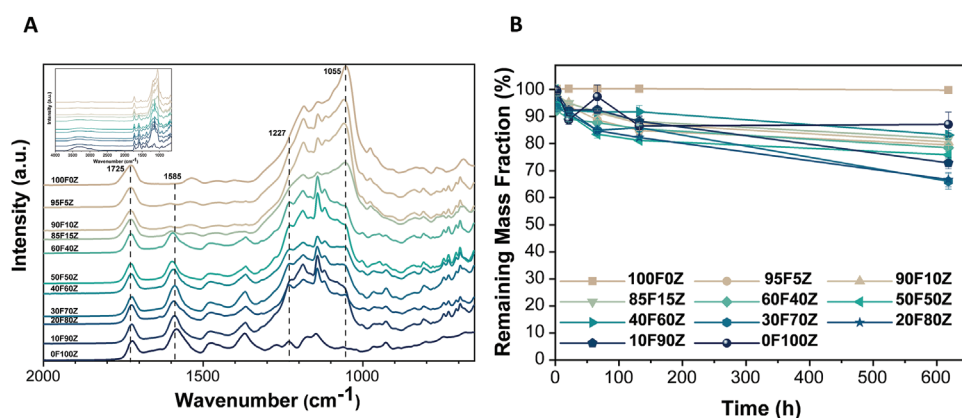


Figure 2. A) ATR-FTIR spectra of the different polymer coatings. Spectra were normalized to the C=O stretching vibration at 1725 cm^{-1} . The incorporation of zwitterionic compound (Z) into the hydrophobic PFPE-matrix (F) is clearly indicated in three vibrational modes: an increase in Z leads to an increase in intensity of the zwitterion's characteristic stretching vibrations $\nu(\text{O-H})$ ($\approx 3300 \text{ cm}^{-1}$) and $\nu(\text{HOC=O})$ (1585 cm^{-1}), and a decrease of PFPE's representative stretching vibration $\nu(\text{C-F})$ (1055 cm^{-1}). B) Stability of the coatings determined as the remaining polymer mass fraction after immersion in Milli-Q water for 2, 21, 66, 132, and 618 h at room temperature compared to the pristine polymer mass prior to immersion.

Table 1. Monomer content, UV exposure, polymerization duration, static and captive bubble water contact angle, and layer thickness (measured via AFM) of the synthesized polymer coatings.

	Monomer unit composition			UV exposure [s]	Polymerization duration [h]	WCA [°]	CBCA [°]	Layer thickness [μm]
	PFPE/CBMA [wt.%/wt.%]	Crosslinker [wt.%]	Initiator [wt.%]					
100F0Z	100/0	3	1	360	16	100 ± 4	65 ± 2	120 ± 1
95F5Z	95/5	3	1	380	16	94 ± 4	34 ± 2	63 ± 1
90F10Z	90/10	3	1	395	16	86 ± 4	27 ± 2	72 ± 1
85F15Z	85/15	3	1	405	16	75 ± 2	26 ± 2	77 ± 1
60F40Z	60/40	3	1	450	16	64 ± 2	15 ± 2	64 ± 1
50F50Z	50/50	3	1	460	16	67 ± 2	12 ± 2	64 ± 1
40F60Z	40/60	3	1	470	16	62 ± 3	11 ± 2	64 ± 1
30F70Z	30/70	3	1	475	16	61 ± 2	11 ± 2	65 ± 1
20F80Z	20/80	3	1	480	16	60 ± 4	14 ± 3	75 ± 1
10F90Z	10/90	3	1	490	16	50 ± 2	10 ± 1	66 ± 1
0F100Z	0/100	3	1	510	16	49 ± 2	8 ± 2	63 ± 1

hydrophilic zwitterionic compound showed unique bands for $\nu(\text{O—H})$ at $\approx 3300\text{ cm}^{-1}$ and $\nu(\text{HOC=O})$ at 1585 cm^{-1} . The change of the modes specific for either compound, i.e., $\nu(\text{O—H})$ and $\nu(\text{HOC=O})$ for CBMA and $\nu(\text{C—F})$ for PFPE, indicated the successful copolymerization of both materials and their presence in the coatings. An increase in CBMA content entailed an increase in the intensity of $\nu(\text{O—H})$ and $\nu(\text{OHC=O})$ together with a decrease of $\nu(\text{C—F})$.

To test the stability of the polymer coatings, the samples were immersed in Milli-Q water for up to 26 days (618 h) and retrieved after certain immersion times (2, 21, 66, and 132 h) to determine the polymer mass gravimetrically on a high precision scale. The remaining mass after the different immersion periods is shown in Figure 2B. After an immersion of ≈ 6 days (132 h) all coatings showed a retained polymer mass of $\geq 82\%$. Even after a further 20 days of water treatment, most coatings exhibited long-term stability of $\geq 75\%$, apart from 50F50Z and 20F80Z, which dropped to $\approx 67\%$ of their original mass. The results for the CBMA-containing samples revealed no correlation between zwitterionic content and coating stability, indicating that the mass loss is not specifically linked to one of the compounds. Overall, all polymers were sufficiently stable to be tested in both the dynamic bacteria and diatom accumulation assays.

Since the ability to interact with water plays a key role in zwitterionic materials, the wettability of the amphiphilic coatings was assessed via sessile water contact angle goniometry (WCA) and captive air bubble contact angle goniometry (CBCA) to reveal the presence of hydrophilic groups at the interface of the coating and their ability to reorient toward the surface underwater to form stable hydration layers. In Table 1 and Figure 3 it is shown that static WCAs decreased with higher zwitterionic content, being 100° for the hydrophobic PFPE coating 100F0Z and 49° for the hydrophilic CBMA coating 0F100Z. For the amphiphilic coatings, the contact angles gradually decreased with higher zwitterionic content: 95F5Z (94°), 90F10Z (86°), 85F15Z (75°), 60F40Z (64°), 50F50Z (67°), 40F60Z (62°), and 30F70Z (61°), and 10F90Z (49°). At CBMA contents ranging from 40 to 80 wt.%, the contact an-

gle seemed to be predominately determined by the hydrophilic CBMA component rather than the hydrophobic PFPE component, discernible from similar WCAs ($63 \pm 3^\circ$) among the different amphiphilic mixtures. At lower CBMA contents of 5–15 wt.%, larger changes in contact angle were observed (94° to 75°). However, since full hydration cannot be reached in air equilibrium, the contact angle was determined by CBCA goniometry underwater (Figure 3A). Without exception, a sharp drop in CAs was observed for all coatings right after immersion relative to the values in air, indicating strong polymer rearrangement. On average, the CA of the coatings decreased by $\approx 30^\circ$ – 40° within already the first minute of immersion. After 10 min, a CBCA of 65° was obtained for the purely hydrophobic polymer 100F0Z, whereas all amphiphilic polymers with CBMA content $> 40\text{ wt.}\%$ quickly revealed hydrophilic CBCAs $< 15^\circ$. For the coating 95F5Z, the contact angle dropped from 95° to 23° immediately after immersion and increased afterward again by $\approx 12^\circ$, which reflects a con-
taphilic behavior.^[66]

As it becomes obvious from Figure 3A,B, the difference between the WCAs and the CBCAs after rearrangement differed for the different polymer compositions. Therefore, the contact angle difference Δ_{CA} between the static WCA and the CBCA after 10 min was calculated for each coating (Figure 3C). The highest rearrangement, meaning largest Δ_{CA} , was observed for 95F5Z ($60^\circ \pm 4^\circ$) which then decreased with increasing zwitterionic content down to $40^\circ \pm 2^\circ$ for 10F90Z. The lowest surface rearrangement was obtained for hydrophobic 100F0Z with a change of contact angle of $35^\circ \pm 4^\circ$. It was surprising to see the highest rearrangement taking place in the hydrophobic samples 95F5Z and 90F10Z ($60^\circ \pm 4^\circ$ and $59^\circ \pm 4^\circ$, respectively). It should also be noted, that 85F15Z and 60F40Z have a comparatively low change in contact angle ($49^\circ \pm 4^\circ$) while 50F50Z, 40F60Z, and 70F30Z are above that value with 50F50Z revealing a local maximum of $55^\circ \pm 3^\circ$.

To evaluate the interaction of the polymer films with water, not only at the surface but also within the bulk material, the water uptake of the coating was determined by gravimetric

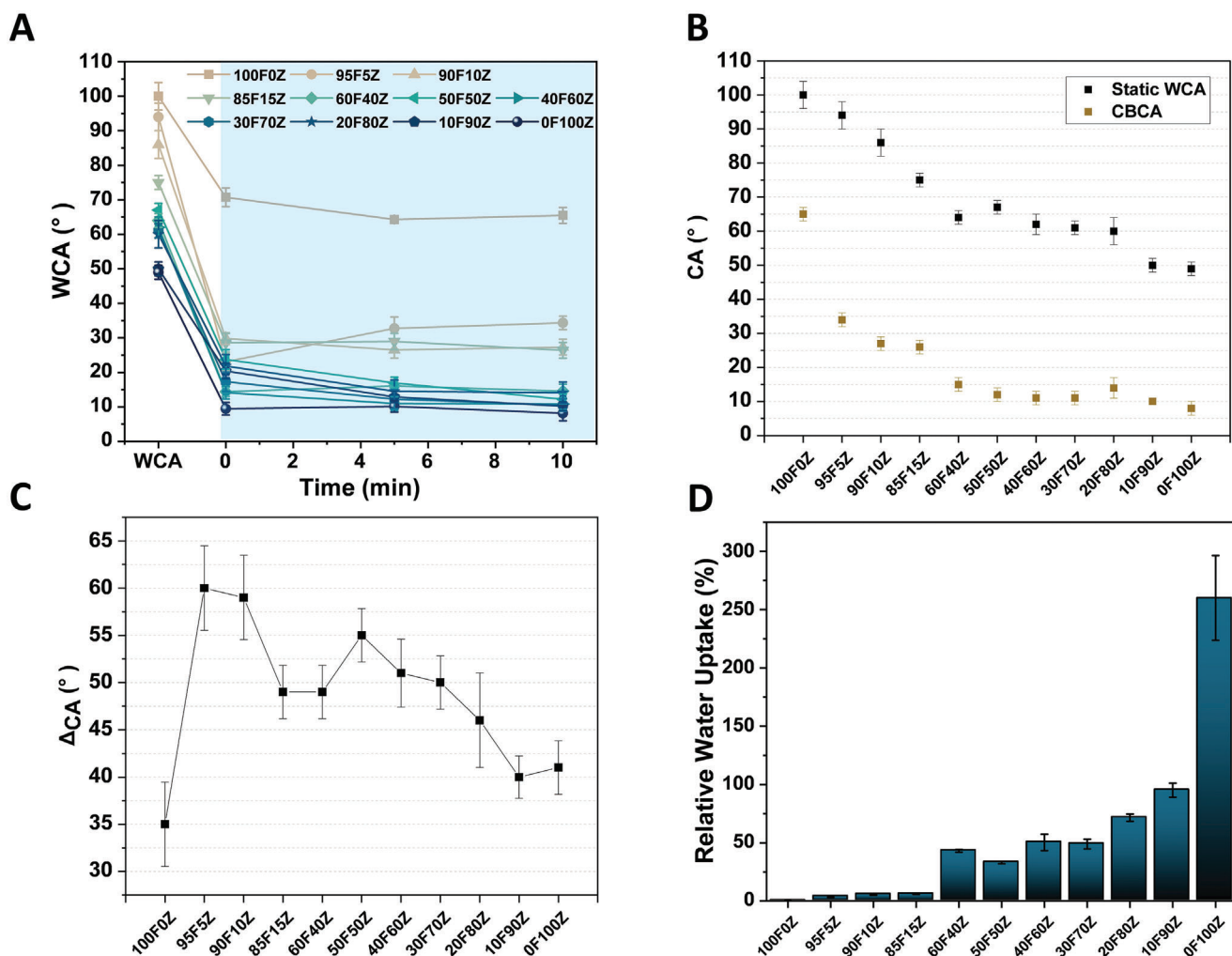


Figure 3. A) CBCA of the polymers with different compositions immersed in water for 10 min. B) C) Difference of WCA and CBCA after 10 min showcasing the surface rearrangement after 10 min immersion. D) Gravimetrically determined relative mass change due to water uptake after immersion in Milli-Q water. Each value represents the mean of three independent measurements with the standard deviation as error bars.

analysis (Figure 3D). The lowest amount of water was found in the most hydrophobic coating 100F0Z (0.15%), while we noticed the highest water uptake for 0F100Z with 260%. In general, higher zwitterionic content in the polymer films led to a higher water uptake. Interestingly, the water uptake in the pure CBMA coating 0F100Z of 260% dropped by more than 60% when incorporating as little as 10 wt.% of PFPE (10F90Z, 95%). This appeared as a rather intriguing result considering that both polymers showed similar wettability in the sessile water contact angle experiment which was 49° for 0F100Z versus 50° for 10F90Z (Table 1) and in the captive bubble measurements which were 8° and 10°, respectively. For all polymers with small zwitterionic content (<40%), a similarly low water uptake is observed, i.e., 95F5Z (4%), 90F10Z (5%), and 85F15Z (6%). For the amphiphilic polymers with similar mass content of PFPE and CBMA (60F40Z, 50F50Z, 40F60Z), the measured water uptake was 33–50% relative to the polymer mass with lowest values of 33% found for the 50F50Z coatings.

Since surface roughness can affect the fouling properties of coatings, the RMS roughness was determined in air and in the presence of salt water by AFM (Figure 4). All hydrophilic and amphiphilic coatings with moderate to high zwitterionic content showed very low surface roughness between 1 and 5 nm. 100F0Z and 95F5Z had rougher surfaces with RMS values of 24 and 20 nm, respectively. After immersion in salt water, the surface roughness decreased to 21 and 11 nm, respectively. The RMS values of the dry amphiphilic coatings decreased with increasing zwitterion concentration from 20 nm for 95F5Z down to 1.4 nm for 0F100Z. For highly zwitterionic coatings with a CBMA content of 60% and above, the RMS values remained between 1 and 2 nm, with the exception of 50F50Z, which showed a slightly higher roughness with an RMS value of 5 nm. In general, all coatings have rather low surface roughnesses smaller than 24 nm, which decrease with increasing zwitterionic content in both, the dry and the wet state. The dry layer thicknesses were found to be 120 μm on the hydrophobic 100F0Z polymer and 63–77 μm for all other coatings (Figure 4B).

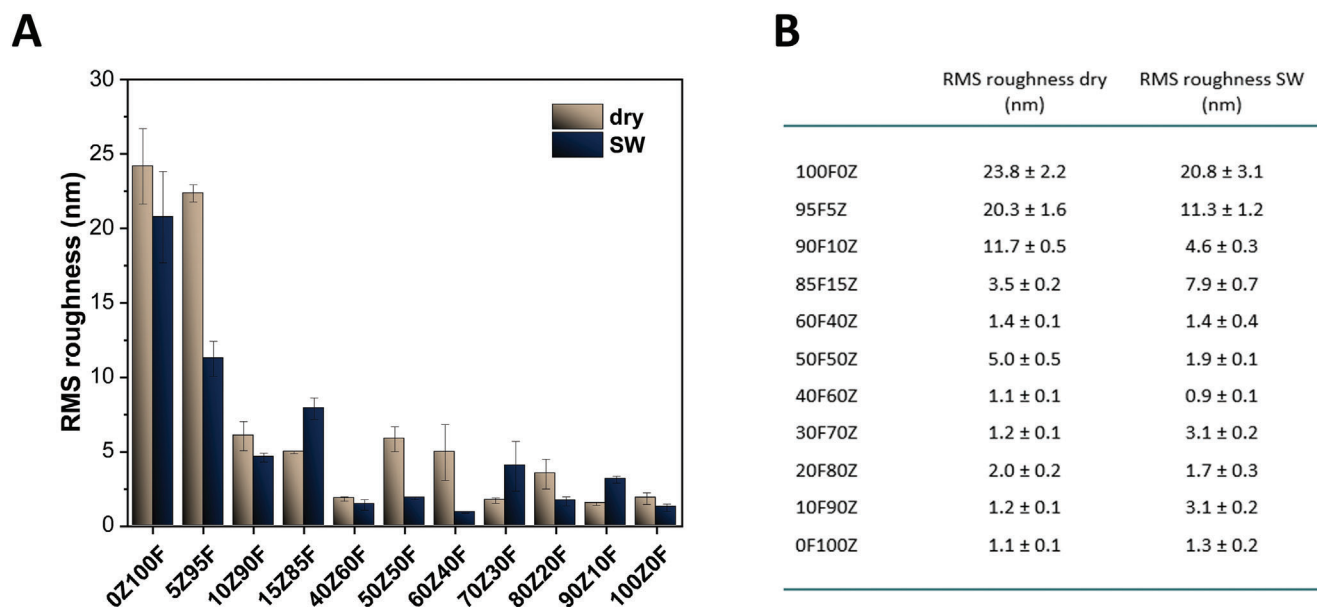


Figure 4. A) Bar chart and B) table of RMS roughness values of dry polymer coatings in the presence of saltwater. Each value represents a mean obtained by three different measured areas of a single AFM image. Error bars show the standard deviation of the mean.

Hydrogels based on PEG or zwitterionic polymers frequently suffer from the incorporation of diatomaceous soil and inorganic particles in the marine environment.^[69] This interaction of the polymer with silt can severely impair the AF and FR performance of coatings in the field.^[67] In order to test the affinity of the polymer coatings to marine sediment, the polymer samples were exposed to a silt suspension on a reciprocating shaker. The amount of absorbed silt was determined by UV/vis spectroscopy as the absorbance $A_{\text{silt}}/A_{\text{pristine}}$ at a wavelength of 440 nm (Figure 5). After an immersion period of 10 min, the hydrophobic polymer 100F0Z (rel. absorbance 1.39) with the least hydrogel character, based on the relative water uptake data, also absorbed the least amount of sand. For the hydrophilic counterpart 0F100Z, the highest uptake of silt was detected, revealing a relative absorbance of 4.39. If the hydrophobic PFPE is blended into the CBMA hydrogel, already 10 wt.% PFPE in the amphiphilic polymer 10F90Z caused a strong reduction (rel. absorbance 2.19) of adsorbed sand to 50% compared to the purely zwitterionic polymer 0F100Z. When CMBA was blended into PFPE, the rel. absorbance increased slowly with increasing hydrophilic content: 95F5Z (1.39), 90F10Z (1.43), 85F15Z (1.85), 60F40Z (2.04), and 50F50Z (3.30). A further increase in zwitterionic mass content above 50% decreased the silt uptake in 40F60Z (2.90), 30F70Z (1.93), and 20F80Z (1.73).

2.1. Dynamic Accumulation Assay of *Navicula perminuta*

The polymer coatings were tested in a dynamic accumulation assay against the marine diatom *Navicula perminuta* to assess the AF and FR properties. Polymeric butyl methacrylate (PBMA), known for its generally high fouling density due to its hydrophobic nature, was included in the experiment as a negative reference. In Figure 6, the adhesion of diatoms onto the polymer

surface of all specimens is presented relative to PBMA. Generally, the highest diatom densities were found on the hydrophobic PBMA reference. The hydrophobic PFPE coating 100F0Z already showed a reduction of 70% compared to BMA with a relative adhesion of 0.30. The hydrophilic counterpart 0F100Z, with ultralow fouling capacity in previous reports,^[67] also revealed a very high resistance against the settlement of diatoms with a relative adhesion of 0.21. When small quantities (increment of

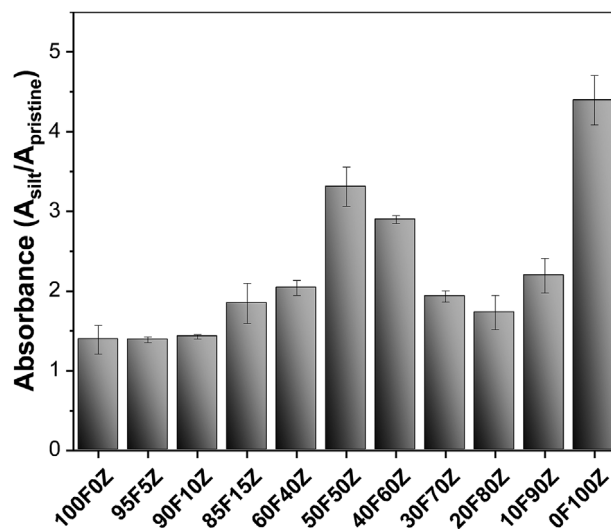


Figure 5. Silt uptake into the polymer matrix as determined by UV/vis spectroscopy. The increase in absorbance after 10 min incubation of the coatings in a silt suspension was measured. The relative absorbance at a wavelength of 440 nm indicates the associated incorporation of sediment. Each value represents the mean of three measurements with the standard deviation as error bars.

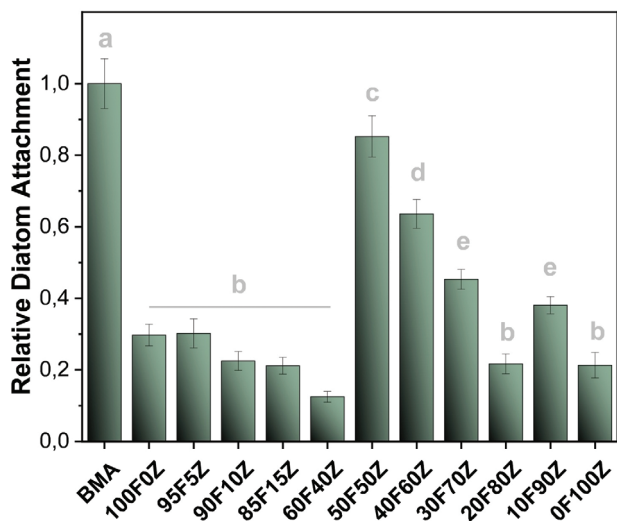


Figure 6. Relative attachment of the diatom species *Navicula perminuta* on polymer coatings relative to PBMA. The samples were preincubated in artificial seawater (ASW) for 30 min and the dynamic diatom accumulation assay was performed for 60 min on an orbital shaker table at 65 rpm with a diatom density of 2 mio mL⁻¹. Each value represents the mean of three independent experiments. Error bars are the standard error of the mean. Total counts were found to be 471, 365, and 230 cells·mm⁻² on the hydrophobic reference PBMA in the three independent experiments. Statistical significance was calculated via an ANOVA with a post hoc Tukey test ($\alpha = 0.05$).

5 wt.%) of zwitterionic CBMA were incorporated into the PFPE matrix, the diatom density decreased from 95F5Z (rel. adhesion 0.30) over 90F10Z (rel. adhesion 0.23) to 85F15Z (rel. adhesion 0.21). This trend, which is decreasing diatom density with increasing zwitterionic content, continues up to the amphiphilic polymer 60F40Z presenting the overall lowest fouling (rel. adhesion 0.13). This translates to a reduction in fouling of 57% toward the purely hydrophobic 100F0Z and 38% toward the purely hydrophilic 0F100Z. The polymer 50F50Z on the other hand, displayed the highest fouling of all investigated CBMA/PFPE polymers with a rel. diatom density of 0.85. The difference of only 10 wt.% CBMA content between 60F40Z and 50F50Z resulted in a 6.5-fold higher attachment for 50F50Z. Then again, the amphiphilic polymers with higher CBMA content appeared to effectively reduce the fouling, which was reflected in a constant decrease of adhered diatoms from 50F50Z (rel. adhesion 0.85), over 40F60Z (rel. adhesion 0.64), to 30F70Z (rel. adhesion 0.45), and to 20F80Z (rel. adhesion 0.21). Thereby, the diatom adhesion on 20F80Z was found to be as low as on the purely zwitterionic polymer 0F100Z (rel. adhesion 0.21). Only the polymer 10F90Z, containing 10% PFPE and 90% CBMA, did not entirely follow this trend and showed a higher diatom accumulation (rel. adhesion 0.38) than 20F80Z. With respect to the statistical significance, purely hydrophobic 100F0Z was performed as well as purely hydrophilic 0F100Z. Among the amphiphilic coatings containing 5–40 wt.% and 80 wt.% CBMA, only 60F40Z (40 wt.% CBMA), revealed a statistically significant difference to 100F0Z. The coatings performed as well as 0F100Z, which is known for its unique ultralow fouling properties.

2.2. Dynamic Accumulation Assay of *Cobetia marina*

The marine bacterium *Cobetia marina* is a biofilm-forming bacterium and is frequently found on submerged surfaces. In recent years it has emerged as suitable model organism to evaluate the resistance of coatings to marine biofouling. Again, PBMA was included in the experiment as a negative control and the results of the investigated repeat assays were normalized on PBMA. The relative bacteria adhesion in Figure 7 followed a very similar trend to the results obtained in the diatom assay. With a relative adhesion of 0.38, the purely hydrophobic 100F0Z polymer showed 62% less fouling than its hydrophobic counterpart PBMA. With the implementation of CBMA into the PFPE matrix, the resulting amphiphilic polymer films 95F5Z (rel. adhesion 0.32), 90F10Z (rel. adhesion 0.28), and 85F15Z (rel. adhesion 0.28) could reduce the bacterial attachment on the coating, which attains its lowest point of fouling with a rel. adhesion of 0.11 in 60F40Z. Then again, incorporating an additional 10 wt.% CBMA to receive an amphiphilic polymer with equal quantities of CBMA and PFPE (50F50Z) led to a severe increase in adhered cells on the surface (rel. adhesion 1.13), even exceeding the fouling of the negative control PBMA. This equals ten-fold higher fouling than on 60F40Z, although the coatings differed only in 10 wt.% CBMA content. Clearly, as observed before, further incorporation of zwitterion could enhance the polymer's AF/FR performance again giving relative diatom densities of 0.54 on 40F60Z, 0.22 on 30F70Z, and 0.13 on 20F80Z. The amphiphilic polymer 10F90Z showed slightly higher fouling than 20F80Z with a rel. adhesion of 0.26. The purely zwitterionic coating 0F100Z resulted in relative bacteria densities as low as 0.11. With regards to

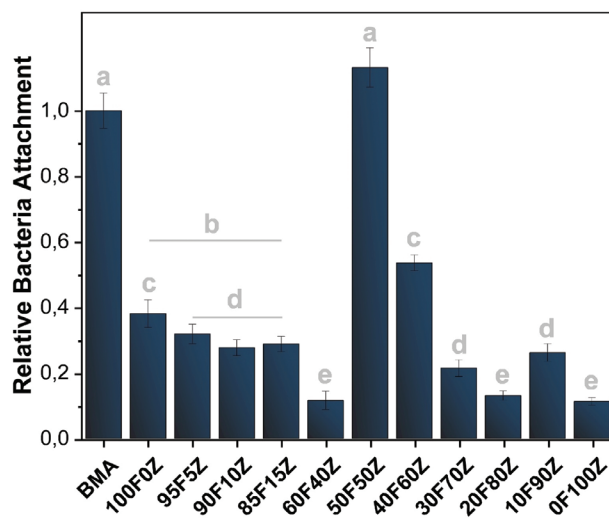


Figure 7. Relative attachment of *Cobetia marina* on polymer samples reported as bacteria attachment relative to PBMA. Samples were preincubated in artificial seawater prior to the 45 min dynamic accumulation bacteria assay on an orbital shaker at 60 rpm with a bacteria density of 7.5 mio mL⁻¹. Each value represents the mean of three independent experiments. Error bars are the standard error of the mean. Absolute bacterial densities were 775, 839, and 471 cells·mm⁻² for the hydrophobic reference samples BMA in the three independent experiments. Statistical significance was calculated via an ANOVA with a post hoc Tukey test ($\alpha = 0.05$).

statistically significant differences, enhanced FR performance toward the purely hydrophobic coating 100F0Z can be reported for the amphiphilic polymers containing 40, 70, and 80 wt.% CBMA. These polymers performed even as well as purely hydrophilic 0F100Z.

3. Discussion

PFPE urethane dimethacrylate and CBMA were in situ copolymerized as transparent, smooth coatings and covalently anchored to silyl-methacrylate functionalized substrates. The immiscible nature of the two monomers can generally entail a polymer pre-assembly step to avoid phase separation.^[63,65,66] This has been overcome by a solvent mixture of TFE and PFO, which was present throughout the entire polymerization process due to the closed reaction chamber. To obtain smooth surfaces relatively high monomer concentration and only small amounts of solvent were used. A similar approach has previously been used to create porous coatings with sufficient amounts and the right choice of solvent.^[59] ATR-FTIR spectroscopy revealed a successful copolymerization of the zwitterionic compound and the hydrophobic PFPE. The increase of the characteristic vibrational modes $\nu(\text{O}-\text{H})$ at 3300 cm^{-1} and $\nu(\text{HOC}=\text{O})$ at 1585 cm^{-1} together with a decrease of the $\nu(\text{C}-\text{F})$ signal at 1055 cm^{-1} provided evidence for the successful copolymerization. In water, the pure PFPE homopolymer 100F0Z demonstrated the least swelling (0.15%), while for the purely zwitterionic polymer 0F100Z, a swelling of 260% was found. The polymers with 5–15 wt.% CBMA revealed only minor changes in swelling with 4%, 5%, and 6% relative water absorption. At high zwitterionic content, the polymers containing 80 and 90 wt.% CBMA exhibited a noticeably lower swelling than 0F100Z with 72% and 95%, respectively, which translates to a 3.6- and 2.7-fold reduction compared to the purely zwitterionic coating. Our data points to a high affinity of CBMA to water, which is in good agreement with previously publications.^[65] Despite the high water uptake, all showed good stability of $\geq 82\%$ within the first 6 days of immersion in salt water and remained very stable ($\geq 75\%$) for up to 26 days.

With rising zwitterionic content, not only the water uptake, but also the sessile static water contact angles (WCA, Figure 3A) decreased from 100° for the hydrophobic 100F0Z to 49° for the hydrophilic 0F100Z. While for larger CBMA contents above 40 wt.% the change in wettability was minor, a strong change was found for CBMA concentrations below this content. This indicates that at higher CBMA concentrations enough hydrophilic groups are present to cause a spreading of the water droplets, while at concentrations ≤ 40 wt.% CBMA, the less hydrophilic PFPE groups increasingly interact with the droplets. For polymers with low zwitterionic contents up to 85F15Z, the swelling was quite low ($< 6\%$) and WCAs from 100° for 100F0Z to 75° for 85F15Z were measured. The swelling (Figure 3D) increased for the polymers 60F40Z, 50F50Z, 40F60Z, and 30F70Z to $\approx 50\%$, with WCA's of 64° , 67° , 62° , and 61° , respectively. Even higher CBMA concentrations led to an enhanced swelling with 70% for 20F80Z, 90% for 10F90Z, and 260% for 0F100Z (Figure 3B). While substantial changes in swelling were observed, hydrophilicity only changed marginally, showcased by their WCA of 60° , 50° , and 49° , respectively. On the basis of this gradual change in water uptake, the similar contact angles of

the three polymers 20F80Z, 10F90Z, and 0F100Z indicate that the surface is largely terminated by the zwitterionic groups while water uptake into the bulk polymer was still hampered at lower CBMA contents, highlighting that even small amounts of hydrophobic PFPE in the polymer network reduce swelling substantially.

As the amphiphilic polymers contain a combination of PFPE with a high water repellency and CBMA with a high affinity to water, the question arose if the polymers can rearrange under water and what kind of interface is subsequently perceived by biofouling marine organisms.^[38,39,50,51] Therefore, the rearrangement was investigated by determining the contact angles in air as well as underwater. While the CBCA of the pure PFPE coatings was found to be 65° , the CBMA containing coatings revealed values between 34° and 8° . All polymer coatings with CBMA contents > 15 wt.% showed CBCAs $\leq 15^\circ$. At lower CBMA contents (5–15 wt.%), the polymers displayed CBCAs between 25° and 35° . In contrast to all other polymers, the 95F5Z polymer revealed a contraphilic behavior.^[68,70–72] After immersion, the contact angle decreased from 95° to 23° , followed by an increase of 12° within 10 min of incubation. The mechanism behind the contraphilic behavior of amphiphilic coatings has been described as an enthalpically driven interaction between the hydrogen atoms of water molecules and the electron-withdrawing groups of the polymer (e.g., nitrogen and oxygen atoms of the urethane group).^[68] Contraphilic wetting was not observed for CBMA concentrations $> 5\%$. Above a concentration of 40 wt.% CBMA, the wettabilities only differed marginally, implying a completely hydrophilic termination at the interface in those amphiphilic polymers. The rapid equilibration suggests unconstrained polymer rearrangement underwater, which is well consistent with a previous report using similar amphiphilic copolymers based on CBMA and DC-PEA which rearranged within only a few minutes.^[67] The rearrangement of surface molecules upon immersion was quantified by calculating Δ_{CA} (Figure 3C), the difference between the WCA and CBCA (Figure 3B). While for the pure PFPE containing 100F0Z, Δ_{CA} was found to be at 35° relatively low, the highest Δ_{CA} values of 60° were found for 95F5Z, the coating with a contraphilic underwater rearrangement behavior. The data indicates that upon immersion a rapid reorientation of the few zwitterionic groups in the polymer toward the surface induces the large Δ_{CA} values. With a slowly progressing water uptake of the PFPE-rich and thus hydrophobically dominated polymer, hydrogen bridges between the zwitterionic moieties might successively form that lead to the burying of part of the zwitterionic groups and expose hydrophobic units instead, leading to the observed contraphilic behavior.⁶⁸ Δ_{CA} seems to decrease with increasing zwitterionic content, reaching the lowest value at 40° for 10F90Z. These results suggest that even amphiphilic coatings with predominantly PFPE contents become hydrophilic upon immersion in water. However, it is rather interesting, that this trend excludes 50F50Z, which has a slightly higher Δ_{CA} than 40F60Z. After 50F50Z, with increasing CBMA concentrations the Δ_{CA} 's decrease again.

Being another physicochemical property relevant to AF performance, we investigated the surface roughness of the coatings by AFM.^[73–77] In general, all polymers revealed very smooth surfaces. The hydrophobic polymer 100F0Z showed the highest surface roughness (24 nm) which was more than 10× rougher than 0F100Z (2 nm). The presence of zwitterionic groups in

the amphiphilic polymer resulted in smoother surfaces which is in accordance with earlier reports.^[67] All dry amphiphilic polymers with a CBMA concentration of more than 15%, except for 50F50Z, had surface roughnesses ≤ 2 nm, which is very smooth considering the total average coating thickness of ≈ 67 μ m. As observed above for the WCA/CBCA difference, the 50F50Z coatings showed a different behavior than all other coatings as its roughness was $\approx 3\times$ as high. For formulations of PFPE with zwitterionic SBMA, also high surface roughnesses were found for pure PFPE coating. Among the smoother amphiphilic SBMA-PFPE coatings, however, a tendency of increasing surface roughness with increasing SBMA content was documented.^[53] For a series of photocrosslinkable polymers also a higher roughness for hydrophobic compositions was reported and assigned to a potential solvent effect.^[34] It was further suggested that the surface roughness underwater, which we suspect to be the more relevant property for marine antifouling efficiency, is a consequence of how the material swells and reorganizes in a specific medium. Here, all coatings with higher CBMA content (>15 wt.%) showed surface roughnesses between 0.9 and 3.1 nm when exposed to salt water, which is a rather negligible difference in surface roughness considering the coatings' average thicknesses of ≈ 67 μ m, indicating that the change in contact angle in the CBCA data is majorly caused by certain freedom of rotation of the polymer chains, e.g. via the so-called crankshaft motion,^[78] to expose hydrophilic groups rather than segregation and reorientation of larger compounds.

Materials with highly hydrophilic properties and high uptake of water, like hydrogels, frequently suffer from the problem of silt absorption in field assays that can enhance the levels of fouling.^[69,79,80] Within only 10 min of immersion in a sand suspension, the coated polymer samples were absorbing severely different amounts of sediment as determined by UV/Vis spectroscopy. The more hydrophobic specimens 100F0Z, 95F5Z, 90F10Z, and 85F15Z embedded distinctly less sand than the polymer 0F100Z with a strong hydrogel character. With higher quantities of incorporated CBMA in the amphiphilic polymers, the uptake of sediment increased. This trend was coherent with the enhanced water uptake of the coatings. The films 20F80Z and 10F90Z exhibited a very low silt absorption despite the high zwitterionic content, with 10F90Z being almost as low as purely hydrophobic 100F0Z. For 10F90Z, silt incorporation was reduced by 50% compared to the purely zwitterionic CBMA coatings, which was surprising as only 10wt% PFPE caused such a strong difference. These findings propose a large impact of small amounts of incorporated PFPE on the silt uptake of the coatings.

Regarding the fouling behavior of the different polymer formulations, we showed that embedding zwitterionic groups in a PFPE-based hydrophobic matrix strongly changed the AF/FR performance of the material. The dynamic attachment assays with *Cobetina marina* and *Navicula perminuta* identified amphiphilic compositions with enhanced but also reduced adhesion resistance relative to the purely hydrophobic (100F0Z) and hydrophilic (0F100Z) compounds. These two chemistries showed already severely lower fouling compared to the negative reference PBMA. This is in good agreement with the findings of Ruiz et al., who noticed a decrease of diatom densities in amphiphilic compositions at 1:1 and 4:1 ratios but at the same time also an increase of fouling in amphiphilic compositions

at 2:1 ratios when comparing SPMA-PFPE polymers to a pure PFPE coating.^[53] This suggests that the incorporation of zwitterions into a hydrophobic matrix does not necessarily improve the AF behavior per se. In our study, it must be noted that the purely PFPE-based coating 100F0Z was already highly resistant to bacteria and diatoms, with FR performance toward diatoms equal to the purely CBMA-based polymer 0F100Z. The fouling on 100F0Z represents a 62% (bacteria), and even 70% (diatom) reduction compared to the corresponding hydrophobic PBMA control. The integration of small zwitterionic quantities of 5, 10, and 15 wt.% improved the fouling reduction in this work, which culminated in the amphiphilic polymer 60F40Z with the lowest relative bacteria (0.11) and diatom (0.13) adhesion among all investigated coatings. The general trend that low hydrophilic contents promote the fouling-resistance is in line with literature reports.^[67] Early studies reported that synergistic effects can be observed when hydrophilic and hydrophobic compounds are combined into amphiphilic networks.^[16,21–24] Both results support the generally observed synergistic effect of amphiphilic coatings and their ability to substantially reduce fouling.

It was surprising though, that increasing CBMA contents led to higher cell attachment on the amphiphilic polymer 50F50Z with a 650% (diatom) and 1000% (bacteria) higher attachment than on the 60F40Z, even exceeding the fouling on the negative control PBMA. A similar observation was made for the silt uptake (+80%) and the surface roughness (+250%) which were both increased for 50F50Z compared to 60F40Z. At even higher CBMA concentrations (40F60Z, 30F70Z, and 20F80Z), fouling was again decreasing with increasing zwitterion content. Both, the diatom and the bacteria assay revealed a difference in accumulation on amphiphilic polymers with low CBMA content, i.e., 5–15 wt.%, and polymers with high CBMA content, e.g., 40 and 80 wt.%, was similarly low. For bacteria however, this difference appeared to be more pronounced with a significantly better performance of the coatings at higher CBMA contents (88% bacteria and 87% diatom reduction for 40 wt.%, 86% bacteria and 79% diatom reduction for 80 wt.%). This suggests that bacteria respond more sensitively to the CBMA content than diatoms, which is again in agreement with the bacteria and diatom experiments in literature, where diatom attachment was found to be slightly higher than bacteria attachment on zwitterionic coatings.^[53]

While a general enhancement of the antifouling performance was revealed for elevated zwitterionic contents, the strong increase in fouling and silt uptake from 60F40Z to 50F50Z was surprising. This sudden increase could not be explained based on hydrophilicity, since similar contact angles and water uptakes were found for the two chemistries.^[69,70] The enhanced surface roughness might support sediment incorporation and biofouling compared to a smoother surface with equal hydrophilic and swelling properties. Besides the high silt uptake, the change Δ_{CA} upon immersion was pronounced for copolymers close to the 50% concentration, indicating a different ability of the coatings to reorganize underwater, which might lead to different surface terminations after water-induced reorganization. These observations might explain the 50F50Z coating with noteworthy characteristics throughout the experiments, setting it apart from the coatings containing less CBMA. The data clearly suggests that specific care has to be taken when formulating such amphiphilic

copolymers, as small changes in zwitterionic content may lead to strong changes in the antifouling performance of the coatings. Besides this interesting observation of the major effects of subtle composition variations, we identified some great performing amphiphilic coatings, both with high and low CBMA content.

4. Conclusion

The hydrophobic PFPE and the hydrophilic carboxybetaines with strongly differing solubility and hydrophobicity were formulated into a series of amphiphilic polymers by photoinduced free-radical polymerization across a broad concentration range and covalently bond to a methacrylate-modified surface using a grafting-through concept. The resulting ultra-smooth and stable polymer coatings were characterized by ATR-FTIR, AFM and contact angle goniometry. When immersed in aqueous media, the polymers revealed quick reorientation of the hydrophilic groups at the water interface. The antifouling and FR performance was assessed against the marine diatom *Navicula perminuta* and the marine bacterium *Cobetia marina* in dynamic accumulation assays. Given the broad content range of the screened polymer blends, it was demonstrated that the performance of amphiphilic copolymers relative to the purely fluorophilic PFPE polymer strongly depended on the individual CBMA/PFPE compositions for both fouling species. Amphiphilic compositions with good antifouling behavior exposed a hydrophilic termination at the interface accompanied by rather moderate water uptake and reduced susceptibility to the incorporation of fouling-enhancing particulate matter compared to pure CBMA coatings. This resulted in amphiphilic coatings with ultralow fouling properties, such as 60F40Z and 20F80Z, where the synergistic effect of hydrophobic and hydrophilic contents had the strongest impact on their marine anti-fouling and FR properties. In turn, this was not observed for coatings with intermediate zwitterion contents (50, 60, and 70 wt.%). While the continuous composition change did not provide a good explanation for the specific performance of dedicated coatings, they were discernible by an unusually large contact angle hysteresis and an increased surface roughness. We demonstrated that the combination of zwitterionic moieties and photocrosslinkable PFPE results in a very attractive amphiphilic material with low fouling, moderate swelling, and concomitantly reduced silt uptake, once the polymer composition is well controlled.

5. Experimental Section

Chemicals: Zwitterionic 3-[[2-(methacryloyloxy)ethyl]dimethylammonio]propionate (CBMA) was purchased from TCI. Perfluoropolyether urethane dimethacrylate (PFPE, Fluorolink MD700) was supplied by Acota. Ethylene glycol diacrylate (EGDA), 2,2-dimethoxy-1,2-diphenylethane-1-one (DMPAP), 3-(trimethoxysilyl)propyl methacrylate (MPTMS), butyl methacrylate and 1H,1H,2H,2H-Perfluoro-1-octanol (PFO) were obtained from Sigma-Aldrich. Glutaraldehyde was purchased from Fisher Scientific, (Heptadecafluoro-1,1,2,2-tetrahydrodecyl)trimethoxysilane (FAS) from Gelest. Ethanol and acetonitrile were acquired from VWR International. 2,2,2-trifluoroethanol (TFE) was supplied by Carl Roth. All chemicals were used as received.

Surface Preparation: Smooth microscope slides (76 × 26 × 1 mm, Paul Marienfeld GmbH & Co. KG, Lauda-Königshofen, Germany) were used as

substrates and chemically functionalized with MPTMS to promote covalent anchoring of the polymer to the substrate. Therefore, the slides were sonicated in ethanol for 3 min, carefully rinsed, dried under a nitrogen stream, and plasma treated for 20 min (0.4 mbar, 80 W, 22 kHz, miniFlecto-PC-MFC, Gala Instruments, Germany) to generate hydroxy groups at the surface. The subsequent MPTMS modification followed a previously published protocol.^[67] In brief, the plasma-treated microscope slides were immersed into a solution of 10% (v/v) of MPTMS in degassed (60 min) toluene. The mixture was sonicated for 16 h at 50 °C and subsequently rinsed in ethanol before allowing the condensation reaction to complete in a vacuum oven for 30 min at 85 °C. The slides were sonicated again for 20 s in ethanol to extract unbound MPTMS and dried in a stream of nitrogen. MPTMS-modified substrates were used immediately or stored under argon.

Preparation of the Polymer Coatings: Photoinitiated free radical polymerization was used to graft polymers to the surface. Monomer solutions of PFPE urethane dimethacrylate and CBMA were mixed in a solvent mixture of TFE and PFO in weight ratios of 0, 5, 10, 15, 40, 50, 60, 70, 80, 90, and 100 wt.% (CBMA/PFPE). It was deliberately decided on a concentration range based on weight percent as the molar weight (PFPE 1880 g·mol⁻¹ and CBMA 229 g·mol⁻¹) strongly differed. Thus, weight percent reflected the ratio of functional carboxybetaine units to tetrafluoroethoxy/difluoromethoxy groups much better. Solvent mixtures were adjusted to the monomer composition and ranged from 50/50 wt.% (PFO/TFE) to 0/100 wt.% (PFO/TFE). For the CBMA and BMA polymers, only TFE was used, while PFPE was prepared from a 50/50 wt.% PFO/TFE solution. As crosslinker 3 wt.% EGDA and 1 wt.% DMPAP as photoinitiator dissolved in acetonitrile were added. The crosslinker content was chosen in a way to obtain a similar monomer:crosslinker density as in pure PFPE, which is also a dimethacrylate but with a relatively long spacer. Thus, with a small error, the crosslinking density throughout the polymer was nearly kept constant. The light-sealed solution was degassed for 45 min before applying ≈250 μL to an airtight, oxygen-free sealed chamber (HybriWell, SecureSeal adhesive chamber, chamber volume 275–320 μL, size L × W × thickness 60 × 21 × 0.15 mm, port diameter 3.2 mm; Grace Bio-Labs, United States) that was reversibly glued to the MPTMS-modified substrates under nitrogen atmosphere. After the solution was pipetted into the chamber, the polymerization was carried out for different irradiation times depending on the mixture ratios between 6.5 and 8.5 min using UV light (UVAcube 100, Dr Hönle AG, 100 W iron-doped mercury vapor lamp, cutoff filter >310 nm, source-sample-distance 18 cm, 15 mW cm⁻²). The UV exposure times were adjusted for each monomer mixture to ensure thorough curing without the coating sticking to the hybrid well chamber. The reaction chamber was kept sealed for another 16 h to allow the polymerization to complete before quenching the reaction by removing the chamber. The polymer films were leached in TFE for 30 min to remove unreacted monomers. The coating was dried in a stream of nitrogen and immediately used or stored under argon.

Attenuated Total Reflection (ATR)-Fourier Transform Infrared (FTIR) spectroscopy: ATR-FTIR spectra were obtained using a Vertex 70 spectrometer (Bruker, Ettlingen, Germany). The spectrometer is equipped with a platinum ATR diamond-crystal and RT-DLaTGS detector. The spectra ranging from 650 to 4000 cm⁻¹ were recorded at a resolution of 4 cm⁻¹.

Stability and Water Uptake Assays: The stability and the swelling of the coatings were determined gravimetrically following previously published protocols.^[67,80] Briefly, the samples were immersed at room temperature for various incubation times (2, 21, 66, 132, and 618 h) in Milli-Q water. After retrieval, water drops on the back side of the glass slide were gently removed prior to weighing the coatings on a high precision scale (METTLER TOLEDO XA105DU, Gießen, Germany). The samples were measured again after a 20 min drying process at 35 °C in a vacuum oven to determine the amount of water absorbed. The weight of the dry samples was compared to the weight of the coatings prior to immersion to characterize the stability of the polymer films in aqueous media for different immersion times. The values displayed represent the mean of three measurements with error bars resembling the standard deviation.

Water Contact Angle Goniometry: The static water contact angle (WCA) and the captive bubble contact angle (CBCA) of the polymer films

were both measured using custom-built goniometers. For the WCA, a droplet of tridistilled water ($\approx 20 \mu\text{L}$) was deposited on the coating and recorded by a charge-coupled device (CCD) camera. Young's equation was used to fit the shape of the water droplet. For the CBCA, downward-facing polymer samples were immersed in a transparent container filled with Milli-Q water prior to dispensing an air bubble onto the surface. The bubble was recorded by a CCD camera and analyzed with the angle tool of the software package ImageJ (version 1.51j) to determine the exterior angle between the surface and the aqueous phase. Each value reports the average of three measurements while the error bars resemble the standard deviation.

Atomic Force Microscopy: $10 \times 10 \mu\text{m}$ topographic images (4 V target amplitude, 2 V setpoint, 750 Hz i-gain, $89 \mu\text{m s}^{-1}$ tip velocity) in air and under salt water (SW) were recorded via atomic force microscopy (AFM) using a NanoWizard 3 (JPK, Germany) equipped with a MULTI75GD-G (Budget Sensors, Bulgaria) and a gold-coated cantilever (spring force constant of 3 N m^{-1}). To quantify the root mean square (RMS) roughness of the surface, the images were analyzed using the software option "statistical quantities" of Gwyddion 2.53. Presented values display the average of three independent analyzed areas with errors being the standard deviation. The thickness of the coatings was obtained by approaching the glass substrate with the cantilever setting the height to zero and approaching the coating with the cantilever on three different spots afterward.

Silt Accumulation Assay: The coatings' affinity to the uptake of silt was determined in a silt accumulation assay as described previously.^[69] Sediment was collected from the seabed at Port Canaveral (FL, USA), autoclaved, and transferred to sterile filtered seawater (CaribSea Pure, Fort Pierce, FL, USA) at a concentration of 37.5 g L^{-1} . To determine the incorporation of sediment into the polymer films, the samples were immersed in the sand suspension for 10 min on a reciprocating shaker at 60 rpm. After retrieval, the slides were washed twice in MQ water for 5 min each to remove salt and unbound silt. Airdried samples were subsequently analyzed via UV/Vis spectroscopy (Agilent, Cary 60 UV-vis, Germany), and the amount of silt was determined via the increase in absorbance at 440 nm.

Dynamic Diatom Accumulation Assay with *Navicula perminuta*: Diatoms (Culture Collection of Marine Algae and Protozoa, Dunberg, Great Britain) were cultivated in sterilized seawater (pH 8.1) containing Guillard f/2 and silica culture medium using a light cabinet (Snijder Scientific, Tilburg, Netherlands) with a light/dark cycle of 12/12 h to mimic a natural day and night cycle.^[81] For the experiment, the nutrient-rich culture medium was exchanged with sterile filtered seawater and sonicated to detach settled cells from the bottom of the flask. Using UV/vis spectroscopy (Agilent, Cary 60 UV/vis, Germany) at a wavelength of 444 nm, the diatom suspension was further diluted until the desired diatom density of 2 mio mL^{-1} was achieved. The polymer coatings were pre-incubated in 4 mL of ASW on an orbital shaker (Neolab, DOS-10L, 65 rpm) to ensure hydration of the films prior to being subjected to the diatom suspension for 60 min at 18°C . After the experiment, the slides were retrieved and treated with 2.5 vol% glutaraldehyde solution to fixate the cells. Prior to microscopic analysis the samples were washed twice in MQ water for 10 min each and airdried overnight in the dark to preserve the auto-fluorescence of the organisms.

Dynamic Bacteria Accumulation Assay with *Cobetia marina*: The original strain (DSM 4741, ATCC 25374) of the marine bacterium *Cobetia marina* was acquired from Deutsche Sammlung von Mikroorganismen und Zellkulturen (Braunschweig, Germany). The cultivation of *Cobetia marina* and preparation for the accumulation experiment followed a recently published protocol.^[82] Bacteria from frozen stocks were streaked onto marine agar plates (sterile marine bouillon (MB) medium with 2% Bacto Agar, Difco, Augsburg, Germany) and grown for one day at 18°C . A single colony was inoculated in 7.5 mL sterile filtered MB medium. The bacterial growth continued on an orbital shaker (Neolab, DOS-10L) at 70 rpm until an optical density of $\text{OD}_{600} \geq 1$ was obtained as measured by UV/vis spectroscopy. The formation of agglomerates could be prevented by filtering and diluting (1/1000) the suspension in a sterile MB medium. The culture was further incubated until an $\text{OD}_{600} = 0.1$ was reached since at this point the bacteria grow in the desired log phase. At this stage, the bacteria suspension was centrifuged (Hettich Zentrifugen, Lauenau, Germany,

3000 rpm, 3 min) and resuspended at a concentration of 15 million bacteria cells mL^{-1} in ASW. The polymers were pre-incubated in 4 mL ASW for 30 min and subsequently mixed with 4 mL of the bacteria suspension to obtain a final concentration of $7.5 \text{ million cells mL}^{-1}$. Analog to the testing with diatoms, the assay was performed on an orbital shaker (60 rpm) for 45 min at 18°C with subsequent fixation and washing steps. For the microscopic analysis, the slides were stained for 30 min with a live/dead staining solution ($1 \mu\text{L mL}^{-1}$ propidium iodide and SYTO 9 in MQ) from Thermo Fisher Scientific (Waltham, USA) and airdried in the dark overnight.

Microscopy and Image Analysis: To quantify the density of attached cells both in the dynamic diatom and the dynamic bacteria assay, the samples were analyzed by fluorescence microscopy. On each sample, a predefined grid of 60 representative fields of view, that covered the entire slide, was automatically scanned by a phase-contrast Nikon Eclipse Ti microscope (10 $\times$ phase-contrast objective CFI Plan Fluor DLL NA 0.3, Nikon, Germany) with an integrated BV-2A filter (excitation: 420 nm; dichroic mirror: 455 nm; barrierfilter: 470 nm; Nikon Instruments) for diatom and Texas Red filter (excitation: 560 nm; dichroic mirror: 495 nm; barrier filter: 630 nm; Nikon Instruments) for bacteria detection. The images were recorded using the software package NIS Elements AR 4.30.02 (Nikon, Germany) with an automated focusing option and subsequent automatic counting. To distinguish between the different chemistries, statistical significance was tested by applying an ANOVA with post hoc Tukey analysis (significance level $\alpha = 0.05$).

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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